

HYBRIDOMA LAB

LAB (6)

Maintenance of Myeloma cells

* Myeloma cells: abnormal lymphocyte

Sp.2

mice origin

characters of sp.2: ① cancerous (divide forever) as long as there are nutrients

② don't produce antibodies

③ HGPRT enzyme (-ve) → blocked

"this enzyme is for pentose pathway"

* Thawing:

transferring cells from -196°

(liquid nitrogen) solid form to

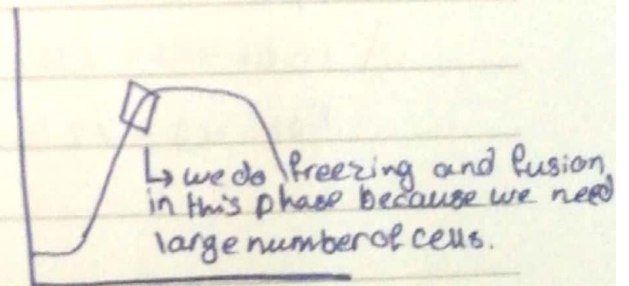
37°C liquid form

caryotub (freezing tube) is a

tube that can tolerate extremely

low temperature

DMSO: a very toxic substance that we add to freeze substances.



* procedure:

1) remove caryotubes from liquid nitrogen jar and immediately thaw them at water bath for 2-3 min. with constant

mixing

6.b — 2) under sterile conditions (Bunsen burner) wash the tube with alcohol

- b.b 3 transfer cells + media to sterile centrifuge tube
- b.b 4 by using a sterile syringe
- 4 add 5 ml 50% F.C.S (media + 50% fcs) drop by drop over 2 min.
- b.b 5 centrifuge at 2500 rpm / 5 min
- b.b 6 At sterile petry dish decant supernatant
- b.b 7 Add 2-5 ml washing media (not ~~not~~ F.C.S) to pellet immediately
- b.b 8 Centrifuge at 2500 rpm / 5 min.
- b.b 9 decant supernatant in sterile petridish
- b.b 10 add 8-10 ml 10% F.C.S to pellet and mix
- b.b 11 transfer cells + media to sterile tissue culture flask and label
- sp2, 10% F.C.S, 6-3-2019, names, group & , section 2
- 12 incubate flask with cells at 5%, 37° incubator for 48-72 hrs.

lymphocyte cell suspension from spleen lab 6

1. Kill the mouse by cervical dislocation or ether anesthesia.
2. Swab the mouse with 70% ethanol and place on right side.
3. Using sterile scissors & forceps, cut the skin in the inguinal region, exposing the peritoneal wall.
4. Cut through the peritoneal wall exposed the spleen. then release spleen from surrounding tissue by forcep.
5. Place the spleen in sterile petri dish containing sterile normal saline (5ml).
6. Transfer spleen into new petri dish containing 5 ml sterile RPMI without FCS.
7. 5 ml ~~sterile~~ syringe fitted with needle is used to take RPMI media and inject it vigorously at multiple sites of spleen [several times] such that spleen cells are seen dispersed.
8. The media containing dispersed cells is transferred to sterile centrifuge tube, cells are centrifuged at 2500 rpm / 5 min.
9. The supernatant is removed & discarded, the cell resuspend into 5 ml of lysing solution tris ammonium (to lyse RBCs) for 5 min then centrifuge at 1500 rpm / 5 min.
- * If the pellet stay red step 9 will repeat again. If not go to step 10.
10. The supernatant is discarded and resuspend the cell into 2 ml RPMI 20% FCS.
- Counting the cells (Adjust to 1×10^7 cell/ml).
- After that make fusion immediately.

Fusion

B-B
 M-B
 M-M

1. After adjusting the lymphocyte cells to 1×10^6 cells/ml, centrifuge at 1500 rpm / 5 min.

2. myeloma cells also centrifuged at 1500 rpm / 5 min.
 1×10^6 cell/ml.

3. Discard the supernatant in both.

4. Resuspend the pellet. ^{then add washing medium} 5ml RPME (-FCS)
 pipettes

5. For each, are resuspended
 Mix the two tubes together and centrifuge them at 1500 rpm / 5 min.

6. Discard the supernatant in Tissue culture 96 well any well manner.

7. Add 0.8 ml of PEG-1500 by a needle slowly
 over 1 min and stand for 1 min. ^{cell membrane} * PEG lyse the cell membrane

8. Add 1ml washing medium RPME (-FCS) by needle over 1 min, then followed by 15ml over 5 min

* Fusion starts

5ml $\rightarrow$ 2 min

5ml $\rightarrow$ 2 min

5ml $\rightarrow$ 1 min

9. Centrifuge at 1500 rpm / 15 min.

10. Discard the supernatant and resuspend gently in 10 ml RPMI 20% FCS

11. Distribute cell mixture between 1x96 well plate (150 wells) 100 μ L per well

12. leave plates in 5% CO_2 , 37 $^{\circ}\text{C}$ overnight.

next days

* In day 2 remove 50 μ L from each well and replace it by ^{add} 100 μ L HAT medium

to kill M-M cells

repeat this process on day 4, 6, 8, 11.

Hybridoma LAB (8) Freezing

* What is Freezing?

changing the media + cells from liquid form to solid form gradually

at room temperature $37^{\circ}\text{C} \rightarrow 0^{\circ} \rightarrow -80^{\circ}\text{C} \rightarrow -196^{\circ}\text{C}$ (liquid N_2)
 $\downarrow$
2 hr - 1 month

* why do we freeze media + cells?

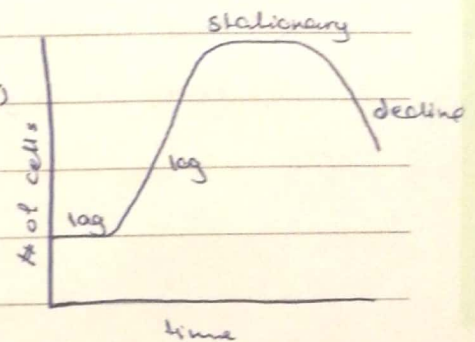
① to decrease loss of cells caused by

contamination (contaminated culture, incubated culture)

② decrease genetic drift

③ prevent senescence الشيخوخة

④ prevent transformation (to cancerous cells)



* when do we freeze?

at late logarithmic phase ($1 \times 10^6 - 1 \times 10^7$ cell/ml) why? to

insure that we have a high population that'll survive the harsh conditions and the chemicals we put during freezing.

* the media used for freezing?

① prokaryotic $\rightarrow$ glycerol

② Eukaryotic $\rightarrow$ D.M.S.O (dimethyl sulfoxide)

* D.M.S.O mixture

46% F.C.S
4% DMSO

40% media
50% F.C.S
10% DMSO

20% Media
20% F.C.S
10% DMSO

نسبة الخلط المناسبة ←

* Function of D.M.S.O

① it works as antifreeze; decrease freezing temperature why?

to avoid forming crystals, thus prevent cell lyses

② keep cells alive

③ decrease temperature in the rate 1°C every $\pm$ min (decrease gradually)

④ DMSO is toxic

* procedure: (under sterile condition)

① take 400-500 μL cells + media to cryotube

② add 400-500 μL F.C.S to cryotube

③ add 100 μL DMSO to cryotube

④ keep cryotube at 0°C for 5-10 min

⑤ transfer cryotubes to -80°C (deep freeze) for 2hr $\rightarrow$ 1 month

Indirect ELISA for the Detection of anti-RBCs Antibodies

ELISA

Enzyme - involves an enzyme reaction as one step in the assay

Linked - the enzyme reaction is directly proportional to the concentration of the unknown substance that is being detected by the assay

Immuno - the assay involves an antibody of known specificity

Sorbent - a reagent is adsorbed to a plastic surface (reaction thus can be "trapped") on the surface

Assay - clinical term for a test to detect the presence of a substance in a test sample

Introduction (read)

The ELISA (Enzyme-Linked Immuno Sorbent Assay) is routinely used for the detection of antibody in an antibody-reaction. The technique also has several other uses, especially in a wide variety of clinical tests where detection of an antibody or a molecule that serves as an antigen is desired.

Our method uses an antibody that has been modified by the addition of an enzyme to the Fc portion (this modified antibody will retain its ability to specifically react with the epitope, and the enzyme remains active as well). Generally, this enzyme-modified antibody is specific for the Fc portion of the antibody that is being tested for in our ELISA (i.e., is an anti-Ig antibody).

Known antigen (in our case RBC's antigens) is adsorbed onto the surface of a special plastic plate designed for ELISA. The test solution (in our case, culture supernatant) is then added. If any antibody of the desired specificity (in our case anti-RBC) is present in the test solution, it will bind to the antigen (that is already adsorbed to the plastic surface).

Then the enzyme-modified antibody is added. If any antibody specific for the antigen is present in the test solution (and thus bound to the adsorbed antigen), the enzyme-modified antibody will bind to the antibody in the test solution. The presence of the enzyme indicates the presence of the anti-Ig, which indicates the presence of the antibody specific for the antigen adsorbed to the plastic surface.

The enzyme can be detected using a known substrate that gives a measurable colored product when acted upon by the enzyme. The generation of a colored product indicates the presence of the enzyme-modified anti-Ig, which indicates the presence of the anti-RBCs (in our ELISA test).

Reagents:

1. Phosphate Buffer Saline
2. Tween20 (looks like soap)
3. Bovine Serum albumin (antigen)
4. ~~Phosphate Buffer Saline~~
5. Stop solution (NaOH and EDTA to neutralize color)
6. ~~Phosphate Buffer Saline~~
7. Your test supernatant (has Ab 1)

o-phenylene diamine
O.P.D. (substrate) (colorless → orange)
Horse radish Peroxidase
(HRP) conjugated with
anti-mouse IgG

+ NaHCO₃
Na₂CO₃ ←
PH < 7

* Passions:
① M-M → will die when adding HAT media

② B-B → will die after 72 hrs

③ B-M 1/3 of cells

* B-M:

① nonsecreted Ab

② secreted non-specific Ab

③ secreted specific Ab

o-u wells of all the plate will be ③

* the secreted Abs will be in the supernatant

* it needs 11 days to start secreting Abs

* BSA not blood

* fusions:

① M-M → will die when adding HAT media

② B-B → will die after 72 hrs

③ B-M $\frac{1}{3}$ of cells

* B-M:

① nonsecreted Ab

② secreted non specific Ab

③ secreted specific Ab

$\frac{1}{9}$
 $\frac{1}{9}$
 $\frac{1}{9}$

0-11 wells of all the plate will be ③

* the secreted Abs will be in the supernatant

* it needs 11 days to start secreting Abs

Hybridoma Lab

ction of
dies

say

concentration of the

ssay

thus can be "trapped" on

ubstance in a test sample

Indirect

ELISA for the anti-RBCs

ELI



*~~Enzyme~~

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the enzyme
linked with Ab2
(anti-mouse IgG)

Linked - the enzyme reaction is directly proportional to the amount of unknown substance that is being detected

Immuno - the assay involves an antibody of known specificity

Sorbent - a reagent is adsorbed to a plastic surface, forming a monolayer on the surface

Assay - clinical term for a test to detect the presence of a substance

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Our method uses an antibody that has been modified with an enzyme (this modified antibody will retain its ability to bind to the antigen and the enzyme remains active as well). Generally, this enzyme is used to detect the presence of the antibody that is being tested for in our test.

Known antigen (in our case RBC's antigens) is adsorbed to a plastic plate designed for ELISA. The test solution (in our case containing the antibody of the desired specificity (in our case anti-RBC)) is added to the antigen (that is already adsorbed to the plastic surface).

Then the enzyme-modified antibody is added. The enzyme-modified antibody binds to the antigen (and thus bound to the adsorbed antigen) and the enzyme binds to the antibody in the test solution. The enzyme then acts on the substrate, which indicates the presence of the antigen on the plastic surface.

*BSA not
blood

*BSA not
blood

The ELISA (Enzyme-Linked Immuno Sorbent Assay) is routinely used for the detection of antibody in an antibody-reaction. The technique also has several other uses, especially in a wide variety of clinical tests where detection of an antibody or a molecule that serves as an antigen is desired.

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The enzyme can be detected using a known substrate that gives a measurable colored product when acted upon by the enzyme. The generation of a colored product indicates the presence of the enzyme-modified anti-Ig, which indicates the presence of the anti-RBCs (in our ELISA test).

Reagents:

1. Phosphate Buffer Saline
2. Tween20 (looks like soap)
3. Bovine Serum albumin (antigen)
- * 4. ~~Phosphate Buffer Saline~~
5. Stop solution (NaOH and EDTA to carbonate buffer pH 10)
6. ~~Alkaline Phosphatase conjugate anti-mouse IgG~~
7. Your test supernatant (has Ab 1)

o-phenylene diamine
O.P.D. (substrate) (colorless → orange)

Horse radish Peroxidase
(HRP) conjugated with
anti-mouse IgG

+NaHCO₃
Na₂CO₃ ←
PH < 7

Procedure:

1. Place 0.1 ml (100 microliters) of the ^{BSA} ~~RBCs~~ ^{phosphate buffer saline} solution (in PBS) in each well of an ELISA microplate. Use one well for each supernate to be tested for the presence of anti-RBCs antibodies.
2. Allow the plate to stand at room temperature for approximately one hour, or overnight at 4°C.
3. Remove the fluid from each well by vigorously flicking the plate and then tapping it upside down onto an absorbent paper surface.
4. Wash with 3X PBS-0.05% Tween-20 by adding 300 microliters each time.
5. Add 0.3 ml (300 microliters) of the blocking solution (3% BSA dissolved in PBS-0.05% Tween-20) to each well (adsorbs to any plastic surface that does not have RBCs covering it, thus preventing any non-specific sticking of antibody to the plastic surface).
6. Allow plate to stand at 37°C for 1 hour (during which the blocking solution will attach to any remaining portion of the plastic surface).
7. Remove the fluid from each well and wash 3X as in step 3 and 4.
8. Using separate pipettes, add 0.05 ml (50 microliters) of negative (fresh tissue culture medium) to two separate wells. These wells will serve as your controls.
9. Add 0.05 ml (50 microliters) of each test supernatant (that has been removed from wells in your hybridoma culture plate using sterile technique) to each of the other (non-control) wells. ~~50-70 min~~ ^{50-70 min} ^{HP} ^{every well needs a different sterile}
10. Allow the plate to stand at 37°C for 1 hour, during which any anti-RBCs antibodies will bind to the RBCs in the test wells. ^(incubator)
11. Remove the fluid from each well and wash 3X as in step 3 and 4.
12. Add 0.1 ml (100 microliters) of AP-conjugate anti-mouse IgG to each well (including controls). ^(enzyme)
13. Allow the plate to stand at 37°C for 1 hour.
14. Remove the fluid from each well and wash 3X as in step 3 and 4.
15. Add 0.1 ml (100 microliters) of substrate (^{OPD} ~~PMP~~) to each well (including controls).
16. Allow the plate to stand at room temperature for 30 minutes.
17. Add 0.1 ml (100 microliters) of stop solution (NaOH and EDTA, PH > 10). ^{stop solution → stop enzyme activity}
18. Read the plate with a microplate reader at 550 nm.

* types of -ve controls:

- ① just ~~washing~~ washing media (pink color)
- ② everything but Ag (no BSA)
- ③ everything but Ab1 (no supernatant)
- ④ everything but Ab2-E
- ⑤ everything but substrate

Procedure:

1. Place ^{0.3} **0.1 ml (100 microliters)** of the ^{BSA} **RBCs** solution (in ^{phosphate buffer saline} PBS) in each well of an ELISA microplate. Use one well for each supernate to be tested for the presence of anti-RBCs antibodies.
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6. Allow plate to stand at 37°C for 1 hour (during which the blocking solution will attach to any remaining portion of the plastic surface)
7. Remove the fluid from each well and wash 3X as in step 3 and 4.
8. Using separate pipettes, add **0.05 ml (50 microliters)** of negative (fresh tissue culture medium) to two separate wells. These wells will serve as your controls.
9. Add **0.05 ml (50 microliters)** of each test supernatant (that has been removed from wells in your hybridoma culture plate using sterile technique) to each of the (non-control) wells ~~phosphate buffer saline~~. (every well needs a pipette)

too less washing: false positive
too many washing: false negative

we wash by first removing the material then washing with PBS 3 times

already done

already done

already done

under sterile conditions

from wells in your hybridoma culture plate using sterile technique) to each of the other (non-control) wells ~~only per assay~~ (every well needs a different dilution)

10. Allow the plate to stand at 37°C ^{50-70 min} for 1 hour, during which any anti-RBCs antibodies will bind to the RBCs in the test wells ^{tip} (→ neutralizer →)

11. Remove the fluid from each well and wash 3X as in step 3 and 4

2nd washing ← 12. Add 0.1 ml (100 microliters) of AP- conjugate anti-mouse IgG to each well (including controls). (enzyme)

13. Allow the plate to stand at 37°C for 1 hour

3rd washing ← 14. Remove the fluid from each well and wash 3X as in step 3 and 4

15. Add 0.1 ml (100 microliters) of substrate (^{OPD} PMP) to each well (including controls)

16. Allow the plate to stand at room temperature for 30 minutes.

DON'T WASH ← 17. Add 0.1 ml (100 microliters) of stop solution (NaOH and EDTA, PH > 10).

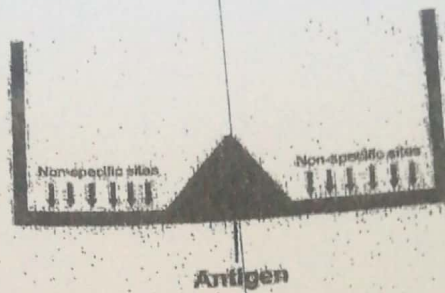
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stop solution → stop enzyme activity

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- ⑤ everything but substrate

1 Antigen Coating



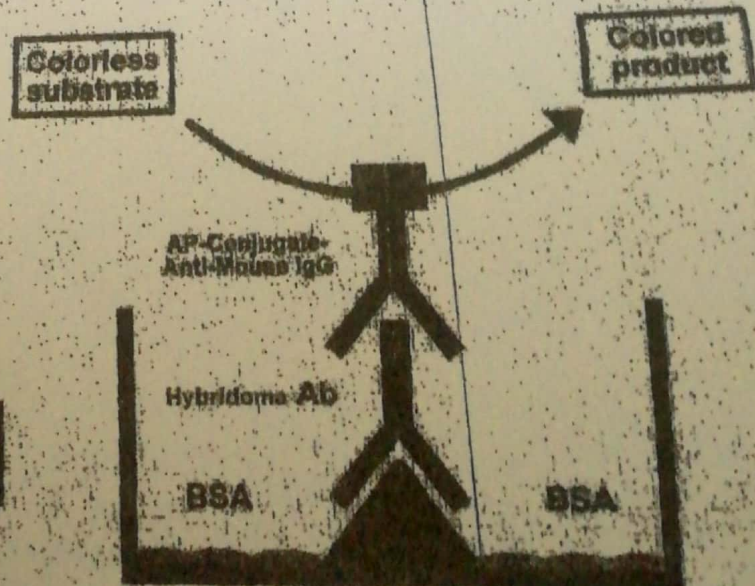
2 Blocking



3 Test Antibody
(your supernatants (Ab's))



QAIA



4 Enzyme-Conjugate Antibody

Hybridoma LAB 10

Feeder cells

* Feeder cells: group of cells that gives nutrients for hybridoma cells for expansion and cloning
expansion: different types of cells that ~~are~~ gives colonies
cloning: one cell that gives a colony

* types of feeder cells:

- ① Macrophages $\rightarrow$ I.P cavity
- ② lymphocyte from spleen $\rightarrow$ splenocyte
- ③ lymphocyte from thymus

* procedure:

- ① Kill mice by ether
- ② swap with 100% alcohol + 70% alcohol
- ③ under sterile conditions take mouse and inject it 3-10 ml sterile P.B.S (F.C.S or washing media) I.P
- ④ massage for 2-5 min to stimulate accumulation of immune cells so the macrophages would stick to the media
- ⑤ by sterile scissor make a small opening in abdominal wall
- ⑥ take fluid from intraperitoneal by sterile pipette and put them in sterile centrifuge tube
- ⑦ centrifuge for 2500 rpm / 5 min
- ⑧ decant supernatant
- ⑨ add 5 ml washing media to pellet and mix gently and centrifuge at 2500 rpm / 5 min
- ⑩ add 1 ml media + 20% F.C.S
- ⑪ count the cells, final concentration should be 1×10^6 cells/ml
- ⑫ take 0.1 ml and add to each well

NO.

Hybridoma (AB II) cloning by limiting dilution

- * clone: group of cells come from one single cell
- * they all should look like each other (circle, edges, ~)
- * limiting dilution: Few dilution times end with one cell in one well
- * 1 well $\rightarrow$ +ve ELISA $\rightarrow$ establish $\rightarrow$ one 96 wells plate
- * practical method:

- 1) add 0.1 ml Feeder cells in all the wells
- 2) take 0.1 ml supernatant from +ve ELISA and put in 1A



- 3) dilute from 1A to 1H (vertically)
 - 4) dilute from 1A to 12A, 1B to 12B, etc (Horizontally)
 - 5) incubate plate at 37°C , 5% CO_2 for 4-7 days
 - 6) make ELISA $\rightarrow$ +ve
- Feeder cells $\rightarrow 1 \times 10^6$ cells/ml / 1×10^5 cells/well
 - if the unit is per well it means it's per 0.1 ml

Cloning by Limiting Dilution

Introduction

Up to now, any desired specific antibody-secreting hybridoma cells have been cultured in a mixed population of cells (including non-secreting hybridomas, cells secreting antibodies of specificities other than the one desired, other cell types that survive in HAT medium, etc.) In this lab, you will separate cells in a mixed population from each other and grow single, isolated cells in culture. The progeny from the cell divisions of each isolated cell will be a clone of identical cells. If the original cell was secreting antibodies, then all of the members of the clone should secrete antibodies that are identical (monoclonal antibodies). Once an antibody-secreting clone has been established, it can be expanded to allow for the production of large amounts of the antibody. Continued growth of this clone will result in a long-term dependable source of monoclonal antibodies.

Cloning by limiting dilution is a procedure for separating cells based on the assumption that if a suspension of cells is diluted with enough culture medium, a concentration of cells will be produced such that an accurately measured volume of the diluted suspension will contain 1 cell. When this volume of the diluted suspension is placed into separate wells of a 96-well plate, each well should receive 1 cell/well. If this cell remains viable (feeder cell layers and/or "conditioned" medium is/are usually needed because of the obviously low cell density of 1 cell/well) and proliferates, then an isolated clone of cells will have been established in the well.

Because the cell counting method used to determine the original cell concentration in the undiluted suspension is based on a rather crude statistical sampling (haemocytometer), not every well will receive 1 cell/well. Many techniques (including this one) also use a suspension of cells that contains 10 cells/ml, and dispense 0.1 ml of this suspension into some of the wells. Thus, some wells may have near 10 cells/well, some will have between 1-10 cells/well, some will have 1 cell/well, and some may receive no cells at all. However, if the procedure of counting and dilution is done carefully, several wells will actually be seeded with a single cell.

Procedure

1. Prepare a suspension of feeder cells as described in an earlier procedure. Adjust the concentration of this suspension to 1×10^6 .
2. Place approximately 0.1 ml of feeder cell suspension into each well of a 96-well plate.

Hybridoma Lec 6 Myeloma cells

* S-cells $\rightarrow$ pro-B $\rightarrow$ pre-B $\rightarrow$ immature B $\rightarrow$ mature B $\rightarrow$ plasma cell (multiple myeloma)

* Myeloma $\rightarrow$ cancerous plasma cell
multiple myeloma $\rightarrow$ cancer patient

* Myeloma + B-cell (that produce Ab) fusion
 $\downarrow$ $\downarrow$
they keep dividing injected mouse

* characteristics of Myeloma cells

1) HGPRT

2) none Ab producing

3) none Ab secreting

* why we don't use B cells alone in tissue culture?
because it will die spontaneously because it's not cancer cell

* the cell produced from fusion $\rightarrow$ Hybridoma cell

* Myeloma cell and B-cell look alike under microscope

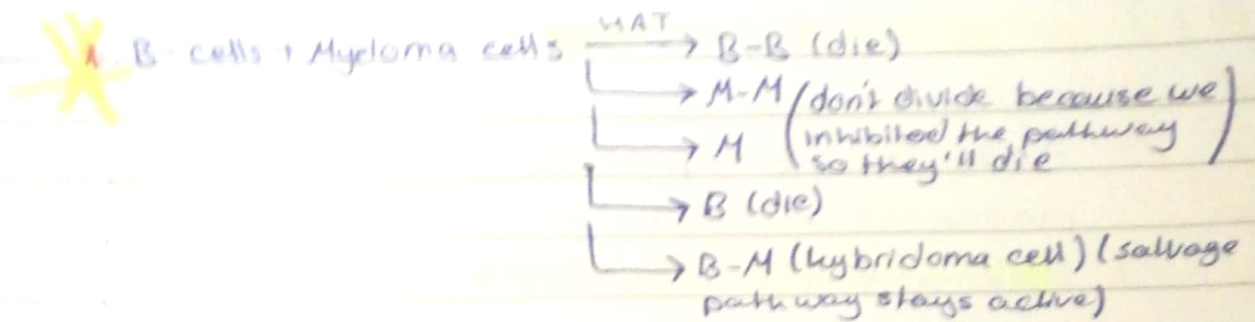
* DNA synthesis $\rightarrow$ denovo (simple atoms H, C, N, O)

$\rightarrow$ salvage pathway (to rescue)

• when we use AMP for synthesis it's considered recycling

• HGPRT $\rightarrow$ recycles

when HGPRT⁻ it means there's a deficiency in that enzyme which makes the DNA unable to do salvage pathway (a mutation by Azaguanine or Thioguanine)



HAT: • H → hypoxanthin

• A → Aminopterin (inhibitor for one of the steps of de novo)

• T → Thymidine (Nitrogen base + sugar)

* when we buy Myeloma it's HGPRT⁻ and this what inhibits the pathway (second one) in M-M (the salvage pathway)

* Maintenance

• 50 ml TC flask

• density $10^5 - 5 \times 10^6$

• every 2-3 months grow myeloma cells in Azaguanine which inhibit DNA synthesis, so the HGPRT⁺ Myeloma cell will die and the HGPRT⁻ myeloma cells will survive

• criteria of Myeloma cell for fusion

log phase → high growth (last division)

density → 10^6 cell

viability → >95% living cell

* Viability test

Trypan blue (stain protein)

selective permeability for stain



shiny white

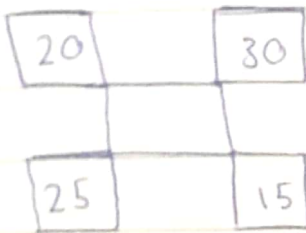
living



blue color

dead

- Counting by hemacytometer



through counting we shouldn't use P.E.S

Average: $20 \times 10^5 = 2 \times 10^5 / \text{ml}$

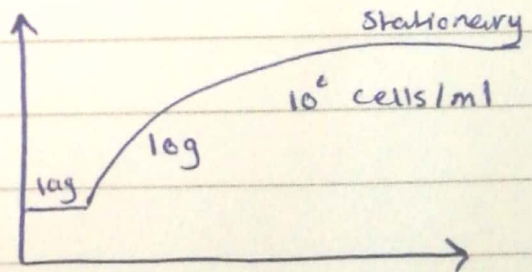
- * characteristic of hybridoma cells:

1) ~~long~~ immortal

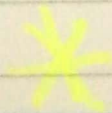
2) produce Abs

- * counting for sp^2 (abnormal lymphocytes) by hemacytometer:

- Freezing of fusion at late log phase, number of sp^2 1×10^6 cells/ml when we can do freezing and fusion 1:10 fusion



- counting by using stain: Trypan blue / nigrosin



Trypan blue (viable stain)

viable
round
shiny

** should be counted
within 5-10 min,
after that all cells
become dead (blue) **

dead
wrinkled
blue

Hybridoma Lec 7

* methods of isolating B-cells from spleen: (mouse sample)

① crushing the spleen using the syringe

② sieving العجبة

a small sieve ~~sio~~ → we put spleen in it → push the spleen into the sieve → cells will be detached in the media why? because lymphocytes are not fixed in the spleen tissues, instead they are floating when we use this method we don't have to centrifuge because the cells are fully detached from the media

③ cutting the spleen into small pieces, then centrifuge (not recommended because cells will not be pure)

* isolation of specific B-cells from human blood sample:

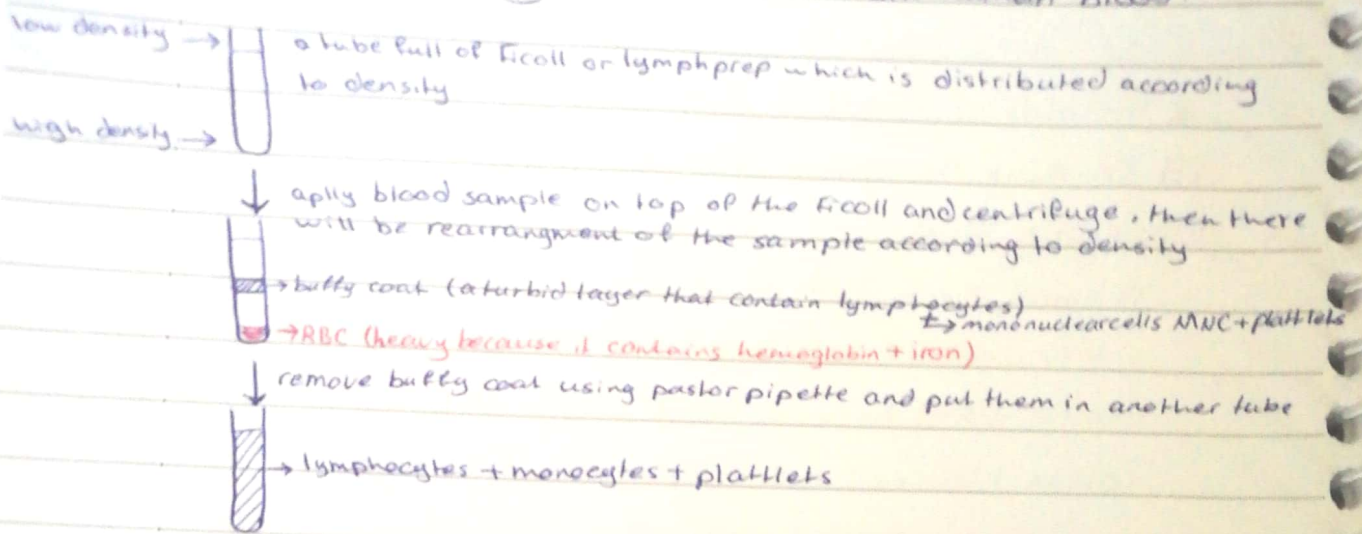
making a monoclonal antibody against human antigen is

very difficult why? ① you can't immunize a human with any antigen you like! (the only case where we can immunize a human with a specific antigen is when we do vaccination)

② we can't take any sample other than blood sample from human, if we want to take a human spleen we must do a major surgery!

③ the percentage of specific monoclonal antibody in human blood is very small (less than 1%)

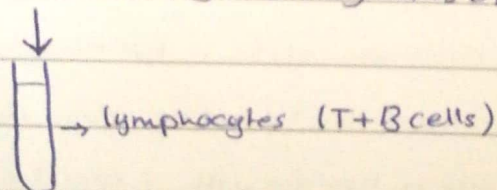
* method of isolating specific B-cell from human blood!



↓ remove monocytes (there are different methods)

- ① add carbon particles
or ② add iron particles

monocytes join phagocytosis for the carbon → gradient centrifugation
→ wash 3 times (the platelets will be removed while washing because they are fragments of cells (very small))



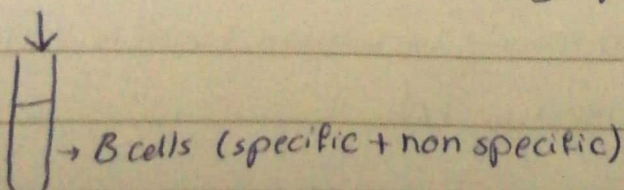
↓ remove T cells (using different methods)

- ① using Nylon wool

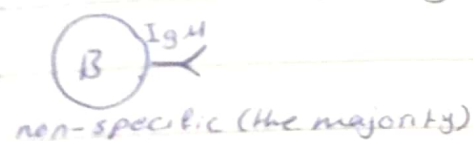
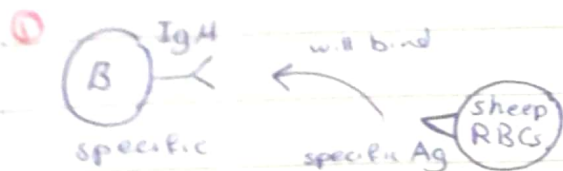
(we put the Nylon wool in a large syringe and pour the lymphocytes and buffer in it, B-cells adhere to the Nylon wool, we detach them using buffer + salts) (up to 95% purity level of B-cells)

- or ② Rosette method

(we add sheep RBCs on the sample that bind to the T cells (only human T cells bind with sheep RBCs) the RBCs and T cells will deposit and B cells will be easily separated)

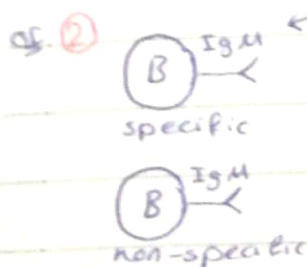


↓ positive selection (isolation of specific B cells)



specific B cell + SRBCs will deposit (density gradient centrifugation)

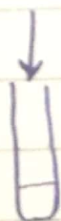
2 B cells has 10^5 IgM on its surface



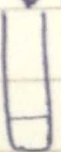
excess specific Ag (something called capping will happen, it's when we add excess antigen until there will be engulfment to the IgM)

↓ then we use Ab against IgM that will bind to the non specific B cells

density gradient centrifugation



↓ transformation (transfer from normal cells to cancer cells) using EBV (epstein bar virus)



- disadvantages:
 - ① dangerous for humans
 - ② dangerous for B-cells
 - ③ it's very difficult to remove the virus from monoclonal Ab thus we can't use it in therapy

Hybridoma Lec 8

* optimal conditions of fusion:

① concentration

the ratio between plasma and myeloma cells should optimally be 1:1 - 10:1 (B:M)

why are B cells more than Myeloma cells? because Myeloma cells are pure while B cells are specific and non specific

* poly ethylene glycole MW 1000-3000 → solution
the one we used was 5000 MW
↓
optimal

PH optimal = 8.2, Temperature upto 37°C

while adding PEG we should shake gently **why?** because PEG disturbed the cell membrane making it unstable, so if we

vigorously shake the solution the cells will be broken

* after adding HAT media some cells die (the unfused M and B cells) and only the fused cells survive, it takes 5 days for all the unfused cells to be dead, the survived hybridoma cells will be fragile and must be treated gently

* expansion: we do it after the fusion, in it we must expand the number of cells **how?** we transfer the cells from the 0.2 ml well to a 2 ml well (the 24-well plate) and use the HT media instead of HAT (HT is without Aminopterin, the DNA synthesis inhibitor) **why?** because now we only have the hybridoma cells and we need to stimulate the proliferation of hybridoma cells. HT has hypoxanthine and thymidine,

hypoxanthine is a substrate for DNA synthesis, and thymidine is important because salvage pathway does not synthesize thymidine so we must add it

* the number of different hybridoma cells depends on the number of the different epitopes we have in the B cells

* the hybridoma cells are polyclonal so far

* we also add growth factors during expansion to enhance the division

growth factors $\rightarrow$ feeder layer (cells that has growth factors isolated from thymus or macrophages from normal mice)

the mice should be 8-12 weeks old, we take their thymus, smash it, take the extract and add it to media, then add them to the hybridoma cells

macrophages were taken from peritoneum (بطن الفأر) because it's the only place where the macrophages are not fixed but floating **how?**

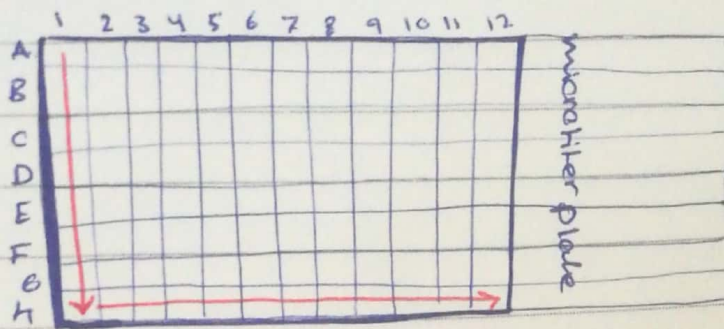
we inject the mouse in its peritoneum with media and massage it then draw the media again

* we do ELISA to test for secretion of Abs, if the hybridoma cells don't secrete Abs there's something wrong

* ~~the~~ how can we separate each clone alone? (cloning is to separate each clone alone) the simplest method is limiting dilution

* the only case where we have 20% concentration of FCS is during fusion * a side note *

* Limiting dilution:



- we dilute until we have only one cell in the last well
- when we find the one cell we culture it
- 3 times dilution usually is enough to find one cell
- when we find the one cell it means we found a hybridoma cell that is specific (secreted one kind of Abs)

* large scale production:

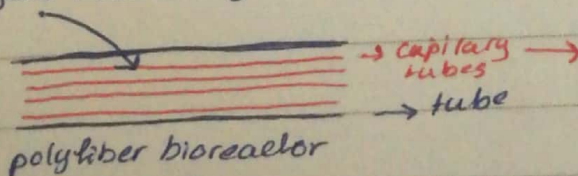
① in vivo : in mice

mouse injected with immune suppressor ^(mineral oil) $\xrightarrow{10 \text{ days}}$ inject the mouse intraperitoneally with the hybridoma cell $\rightarrow$ hybridoma cell (cancerous) will divide and secrete monoclonal antibodies $\xrightarrow{2 \text{ weeks}}$ we take a sample $\rightarrow$ centrifuge $\rightarrow$ take the Ab

- disadvantage: a tumor is induced in the mouse
- the mouse here act as an incubator for the hybridoma cell

② in vitro : bioreactor

inject media + hybridoma cell



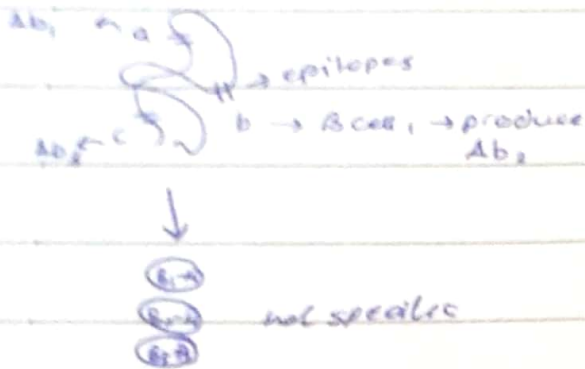
small spaces that allows the media to go in and out but not the cells

$\rightarrow$ then put it in the incubator to allow cells to divide and produce clones

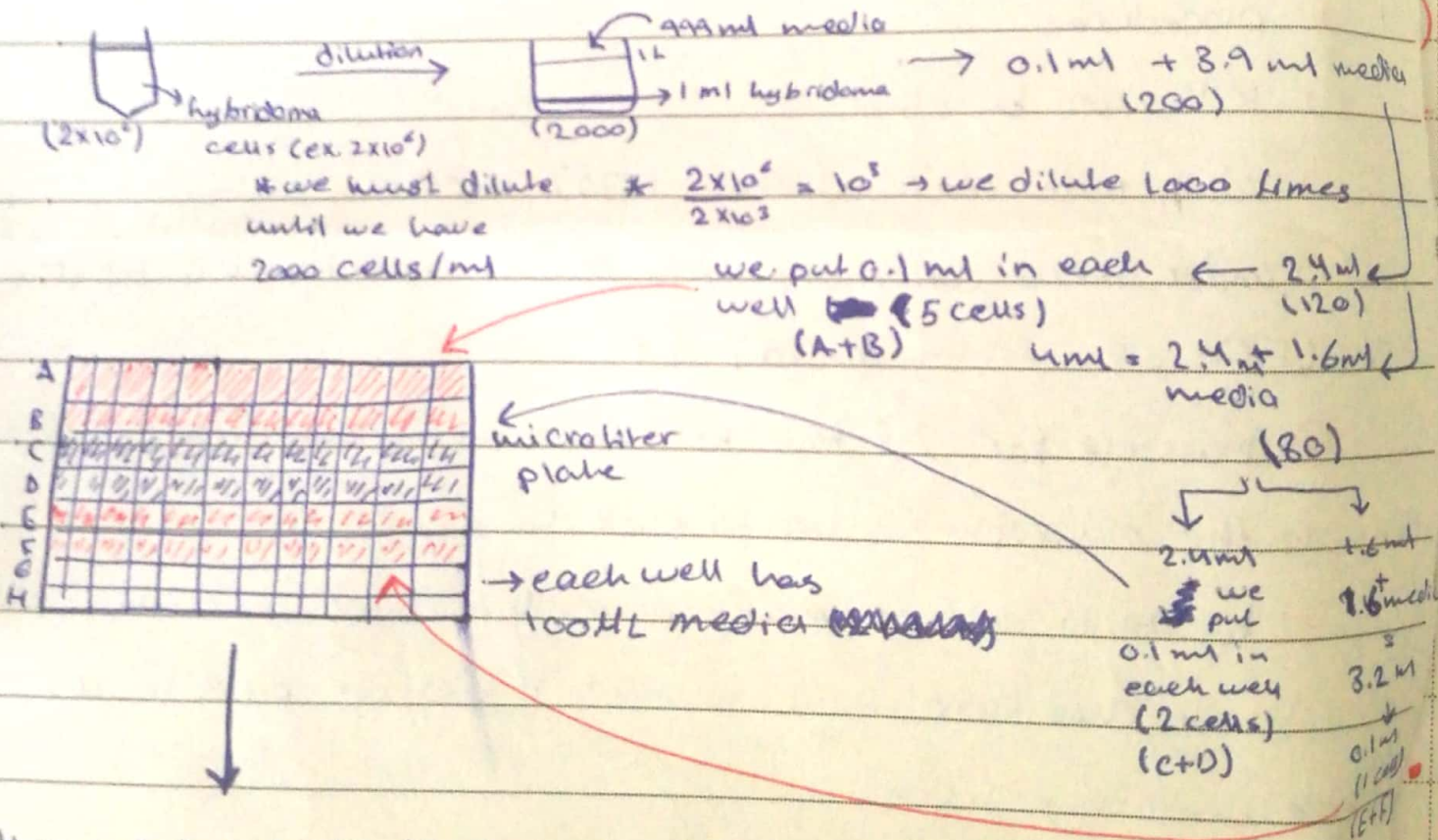
the bioreactor should be used only for one specific cell

Lec 9 hybridoma

A. cloning



Dilution



then we incubate it to allow it to clone.

Hybridoma Lec 10

* characterization: procedures that we do to know the characters of the ~~Ab~~ monoclonal Ab we produced in the potentials of Advertisment

* characterisation methods: (what we use to characterize)

- 1) titer
- 2) isotype
- 3) affinity
- 4) specificity
- 5) purity
- 6) titre

* ~~titre~~ specificity

- making sure the Ab is specific (react with one Ag) using ELISA
- the Ab should react 100% with one Ag

* purity

- we use electrophoresis (we should come out with 2 bands (heavy chain + light chain))

* titer

- concentration
- we use ELISA (we put specific conc. in each well and then do serial dilution for the Ab) then we get a color (which means that indeed there is an Ab)
- the lowest dilution that give positive results $\rightarrow \frac{1}{1000}$
- 5000 titer dilutions is stronger than 1000 titer dilution because it means that if we diluted 5000 times we'll still have positive results
- titer = the activity of the antibody

* isotype

• we want to know if the Ab is IgG/IgM/IgD/IgA

• we use ELISA.

Ab against Gamma 1 (which only exist on IgG) $\xrightarrow{\text{color (it's IgG!)}}$ it's not IgG!
 $\xrightarrow{\text{no color}}$

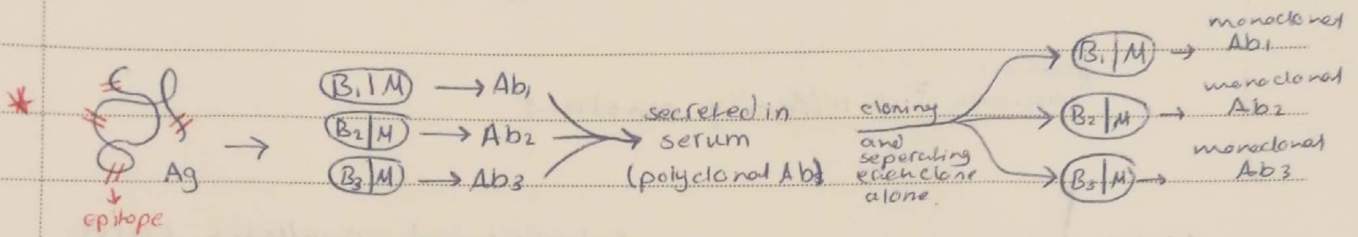
Gamma 2 " " " " IgG2

for the 4 classes and subclasses until we know the isotype

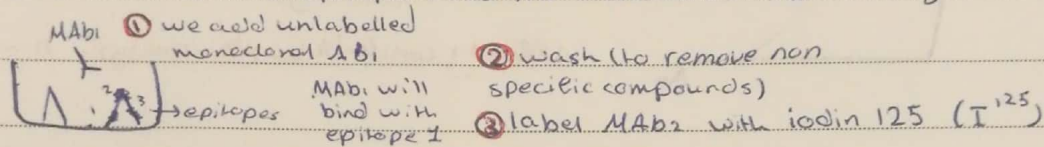
• we do that in one plate, every isotype is tested in two wells

• $\psi_1 \rightarrow \text{IgG1}$, $\psi_2 \rightarrow \text{IgG2}$, $\psi_3 \rightarrow \text{IgG3}$, $\psi_4 \rightarrow \text{IgG4}$, $\alpha_1 \rightarrow \text{IgA1}$,
 $\alpha_2 \rightarrow \text{IgA2}$, $M \rightarrow \text{IgM}$

Hybridoma Lec II



* how do we test which epitope does the monoclonal Ab belong to?



④ wash

MAb₂-I¹²⁵

⑤ inspect with gamma counter if $\rightarrow$ radioactivity detected $\rightarrow$ identical
 $\rightarrow$ no radioactivity detected $\rightarrow$ non identical

(what we inspect with gamma counter is the washing solution)

if there was radioactivity detected, it means that the Ab₂ is identical to Ab₁ why? because at first when we added the unlabelled ~~MAb₁~~ Ab₁ all the epitopes bound with the Ab₁, after that when we added the labelled Ab it didn't bind with any epitope because there was none left, which means that Ab₁ and Ab₂ are the same.

we ~~choose~~ choose the epitopes based on the relatedness (related proteins / hormones, etc.)

* testing for affinity (binding strength) (sensitivity)

affinity is not connected with speed of binding

affinity test:

① in microtiter plate, we do serial dilution for the Ag

② add labelled monoclonal Ab (MAb-I¹²⁵)

when affinity is low $\rightarrow$ we have 1000 Ag and 1000 Ab

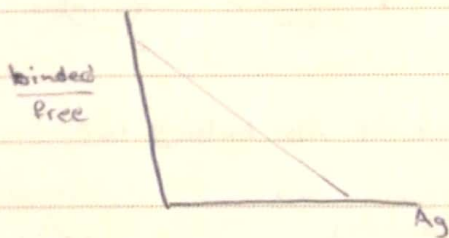
in case of low affinity only 100 (for

example) Ab-Ag complex will be produced

and the rest will be free Abs (900)

when affinity is high $\rightarrow$ all the 1000 Ag and 1000 Ab will combine

$$\frac{\text{products}}{\text{reactants}} = K \text{ affinity constant}$$



Schatcherd analysis curve
for affinity

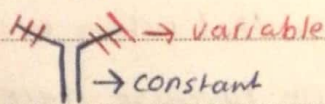
affinity constant) is also analysis) 11 units 151

* monoclonal Abs are used as therapeutic agents ex.

herceptin: a monoclonal Ab used to treat breast cancer

- herceptin when used in its pure form cause dangers on the patients such as inflammation, so there was about 20%.

healing though, so how can we prevent the negative reaction?



the constant part is what recognised mostly as foreign (the variable is also recognized as foreign but less than the constant - about 25%.)

so now, we can't change the variable because the specificity will change, so we must change the constant from mouse to human **how?** by changing the genes, making it chimeric. that made the positive effect of the Ab up to 60%, but the inflammation and the fever didn't go away, so we changed the semi constant (relatively constant) ~~hyper variable~~ parts human as well

